

The University of Lahore
Department of Computer Science & IT
Midterm Examination



Summer-2022

Course Title:	Programming Fundamentals	Course Code:	CS020102	Credit Hours:	4
Course Instructors:	Dr. Abbas Khalid, Ms. Naila Rafique	Program Name:	BSCS		
Time Allowed:	90 Minutes	Maximum Marks:	40		
Date:	16-07-18	Course Mentor:	Dr. Abbas Khalid		
Student's Name:		Signature:			
Reg. No:		Section:			

Objective Part (overwriting and cutting is not allowed)

Time: 15 Minutes

Question # 1: Circle the correct option.

(10 Marks)

1.	C programs are converted into machine language with the help of a) An Editor b) A Compiler c) An Operating system d) none of them
2.	In the following expression $z = x + y$, x and y are called: a) Operators b) Literals c) Values d) Operands
3.	Which of the following can replace a simple if-else construct? a) While loop b) Ternary Operator c) do-while loop d) for loop
4.	Which one of the following is NOT a valid identifier? a) var_abc b) _varabc c) varabc1 d) 1varabc
5.	AND, OR and NOT are logical operators. What data type is expected for their operands? a) Integer b) Boolean c) Decimal d) Character
6.	The break statement causes an exit a) From the innermost loop only b) From all loops & switches. c) Only from the innermost switch. d) From the innermost loop or switch.
7.	The value of a in : $a = (b=3, b+2);$ a) 5 b) 3 c) 2 d) 4
8.	for (; ;) a) is not valid b) means the test which is done using some expression is always true c) will loop forever d) none of them
9.	Which of the following must be present in switch construct? a) case followed by value b) default c) Expression in () after switch d) all of these
10.	In if conditional control, more than one sequential statement can exist between each statement part. a) TRUE b) FALSE

The University of Lahore
Department of Computer Science & IT
Midterm Examination

Summer-2022

CS020102-Programming Fundamentals

Subjective Part	Time: 75 Minutes	Total Marks: 30
------------------------	-------------------------	------------------------

Question # 2: Sow output or identify errors if any	(4*5=20 marks)
---	-----------------------

CODE	OUTPUT
<pre>#include <iostream> using namespace std; int main() { int a = 10, b=22, c=2, x; x = a > b > c; cout<<"The result of the first expression is = "<< x <<endl; x = a < b < c; cout<<"The result of the second expression is = "<<x <<endl; return 0; }</pre>	The result of first expression is = 0 The result of first expression is = 1
<pre>#include<iostream> using namespace std; int main() { for(int j = 1; j <= 4; j++){ for(int k = 1; k <= 3; k++){ if(j == 2&& k == 2){ break; } } cout << j << " " << k << "\n"; } } }</pre>	<pre>1 1 1 2 1 3 2 1 3 1 3 2 3 3 4 1 4 2 4 3</pre>
<pre>int n = 1, i = 1; do{ cout << n << " "; while (i <= 10) { if (i < 5 && i != 2) cout <<"X"<<" "; i++; } }while (n++ <= 3);</pre>	1 X X X 2 3 4

<pre>#include <iostream> using namespace std; int main() { int a = 10; int b = 30; int max = a > b ? a : b; cout << "Maximum Value: " << max << "\n"; return 0; }</pre>	Maximum Value: 30
--	--------------------------

Question # 3: Code the following: (10 Marks)

1. Use the switch statement to ask user to input number of week's day (1-7) and translate to its equivalent name of the day of the week (e.g., 1 to Sunday, 2 to Monday,, 7 to Saturday).

Sample Output Enter the number of week's day: 4 Wednesday
--

Sample Output Enter the number of week's day: 10 You entered wrong number of day

Solution:

```
#include<iostream>
using namespace std;
void main()
{
    int dow;
    cout<<"Enter number of week's day (1-7): ";
    cin>>dow;
    switch(dow)
    {
        case 1 : cout<<"\nSunday";
            break;
        case 2 : cout<<"\nMonday";
            break;
        case 3 : cout<<"\nTuesday";
            break;
        case 4 : cout<<"\nWednesday";
            break;
        case 5 : cout<<"\nThursday";
            break;
        case 6 : cout<<"\nFriday";
            break;
        case 7 : cout<<"\nSaturday";
```

```
        break;
    default : cout<<"\n You entered wrong number of day";
        break;
    }
    getch();
}
```

2. Use nested loops to construct a program that displays a pyramid of Xs on the screen. The pyramid should look like this.

Sample Output

```

    X
   XXX
  XXXXX
 XXXXXXX
XXXXXXXXX
```

Solution:

```
#include<iostream>
using namespace std;
int main()
{
    int i,j,k;
    for( i=1 ; i<=5 ; i++)
    {
        for ( j=1 ; j<=5-i ; j++)
        {
            cout<<" ";
        }
        for ( k=1 ; k<=((i*2)-1) ; k++)
        {
            cout<<"X";
        }
        cout<<endl;
    }
    return 0;
}
```